Reading the First 100 Bytes over Serial

Giga-tronics 8650 / 8600A service monitor - manual procedure with HyperTerminal (Windows)
Target: the first 100 bytes of `8600_155.bic` - revised 13 Oct 2026

Revision 13 Oct 2026. The first edition used the addresses 0x6010 / 0x6038 / 0x6060. The image actually sits at **0x6000**, so the correct commands are `PEH06000`, `PEH06028` and `PEH06050`. The expected replies are unchanged. This edition also explains the power-on window in which the monitor accepts `*`.

What this does

The service monitor's `PEH` ("Peek, Hex") command reads bytes from a memory address and returns them as hex text. On the running instrument the firmware image is mapped at **runtime = file offset + 0x6000**, so the start of the `.bic` file is read at address **0x6000**. Each `PEH0` command returns 40 bytes, so three commands cover 120 bytes - you keep the first 100.

1. HyperTerminal connection settings

Setting	Value
Connect using	the COM port your serial cable is on (e.g. COM1)
Bits per second	9600 (from the firmware's UART set-up; if no reply, retry at 19200, then 115200)
Data bits	8
Parity	None
Stop bits	1
Flow control	None

Then open **File > Properties > Settings > ASCII Setup** and set:

- **Echo typed characters locally**: on. The monitor does not echo, so this lets you see what you type.
- **Send line ends with line feeds**: off. Enter sends a carriage return, which the monitor wants.
- **Append line feeds to incoming line ends**: optional, for tidier display.

2. Commands to send

Type each line exactly as shown. The commands are not case-sensitive. Press **Enter** at the end of each `PEH` line; that sends the carriage return the monitor needs. Do *not* press Enter after the `*`.

Step 1 - wake the monitor during power-on.

The monitor only listens for `*` for a short time (estimated 1-2 s) after the power-on self-test; after that the normal instrument software starts. So:

1. Switch the instrument off and open the HyperTerminal connection.
2. Switch the instrument on and keep pressing `*` (about every half second) until a reply appears.

The instrument should reply:

```
OK 8600A
```

If it replies `ERR 8600A`, a self-test flag is set but reads still work. If nothing comes back and the instrument boots normally, the baud rate or cable is wrong (a null-modem cable may be needed). Switch off and try again. Extra `*` characters typed after the banner are ignored.

Step 2 - read bytes 1-40. Type this and press Enter:

```
PEH06000
```

Reply (40 bytes = 80 hex characters):

```
002D8FFC00009D2A000095F60000961600009656000096360000911800009 11C0000912000009124
```

Step 3 - read bytes 41-80. Type this and press Enter:

```
PEH06028
```

Reply:

```
000096760000969600009130000091340000913800009 13C000091400000914400009 14800009 14C
```

Step 4 - read bytes 81-120. Type this and press Enter:

```
PEH06050
```

Reply (you only need the first 40 hex characters = bytes 81-100; the rest is extra):

```
000091500000915400009158000091 5C000091600000952800009168000091 6C0000917000009174
```

3. Command reference

Type this	Reads address	File offset	Bytes
`PEH06000`	0x6000 - 0x6027	0x00 - 0x27	1 - 40
`PEH06028`	0x6028 - 0x604F	0x28 - 0x4F	41 - 80
`PEH06050`	0x6050 - 0x6077	0x50 - 0x77	81 - 120 (keep 81-100)

Command format: `PEH` + one count digit + address. The count digit `0` means the default 40 bytes, and `1`-`F` read 1-15 bytes. The address that follows is in hex. So `PEH06028` = "peek 40 bytes at 0x6028".

4. Putting the 100 bytes together

Concatenate the three replies and take the first **200 hex characters** (= 100 bytes). The result is:

```
002D8FFC 00009D2A 000095F6 00009616 00009656 00009636
00009118 0000911C 00009120 00009124 00009676 00009696
00009130 00009134 00009138 0000913C 00009140 00009144
00009148 0000914C 00009150 00009154 00009158 0000915C
00009160
```

Quick sanity check: the very first 8 bytes must be `00 2D 8F FC 00 00 9D 2A` - the initial stack pointer (0x002D8FFC) and reset program counter (0x00009D2A) of the firmware. If you see those, the link, baud and address are all correct. If `PEH06000` returns `0000965600009636...` instead, you are reading 16 bytes too far (address 0x6010).

5. Notes

- These bytes are a table of 32-bit big-endian values (the CPU exception vectors; the firmware sets VBR = 0x6000), which is why the first 100 bytes look repetitive.
- To read file offset X of the `.bic`, peek address 0x6000 + X. Address 0x0 holds the boot block, which is not part of the `.bic`. The data block at 0x4000 (probably calibration data) is not part of it either.
- The monitor keeps accepting commands until power-off. You only need the `*` once per power-on.
- For a full automated dump rather than this manual read, use the `gt8650_dump.py` tool. Start it first and then switch the instrument on; it sends `*` by itself.